Module V
Machine Translation And Applications

Dr.S.Periyanayagi
Associate Professor/ SCAI

Contents

- Machine Translation - Introduction
- Basic Issues in Machine Translation
- Statistical Translation
  - Word Alignment
  - Phrase based Translation
  - Synchronous Grammars

Machine Translation

- **Machine translation (MT)** is a domain of computational linguistics, which explores the use of software to translate text or speech from language to another.
- Machine translation simply performs substitution of words in one language for words in other, but that may not assure good translation.

What is Machine Translation?

Automatic conversion of text/speech from one natural language to another

Be the change you want to see in the world

वह परिवर्तन बनो जो संसार में देखना चाहते हो

Machine Translation Usecases

Government

- Administrative requirements
- Education, Security

Enterprise

- Product manuals
- Customer support

Social

- Travel (signboards, food)
- Entertainment (books, movies, videos)

Translation under the hood

- Cross-lingual Search
- Cross-lingual Summarization
- Building multilingual dictionaries

Why should you study Machine Translation?

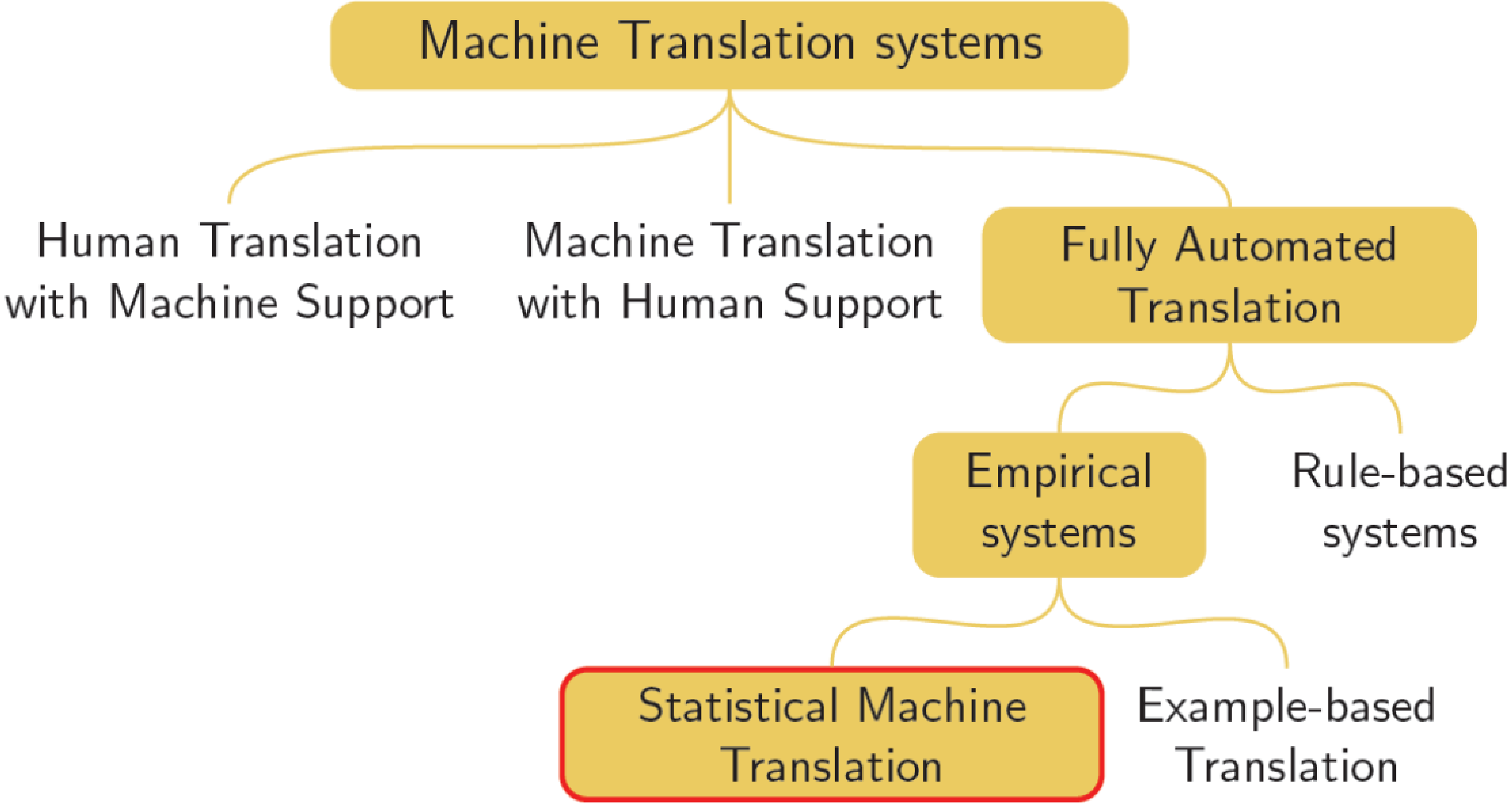
- One of the most challenging problems in Natural Language Processing
- Pushes the boundaries of NLP
- Involves analysis as well as synthesis
- Involves all layers of NLP: morphology, syntax, semantics, pragmatics, discourse
- Theory and techniques in MT are applicable to a wide range of other problems like transliteration, speech recognition and synthesis

Why is Machine Translation interesting?

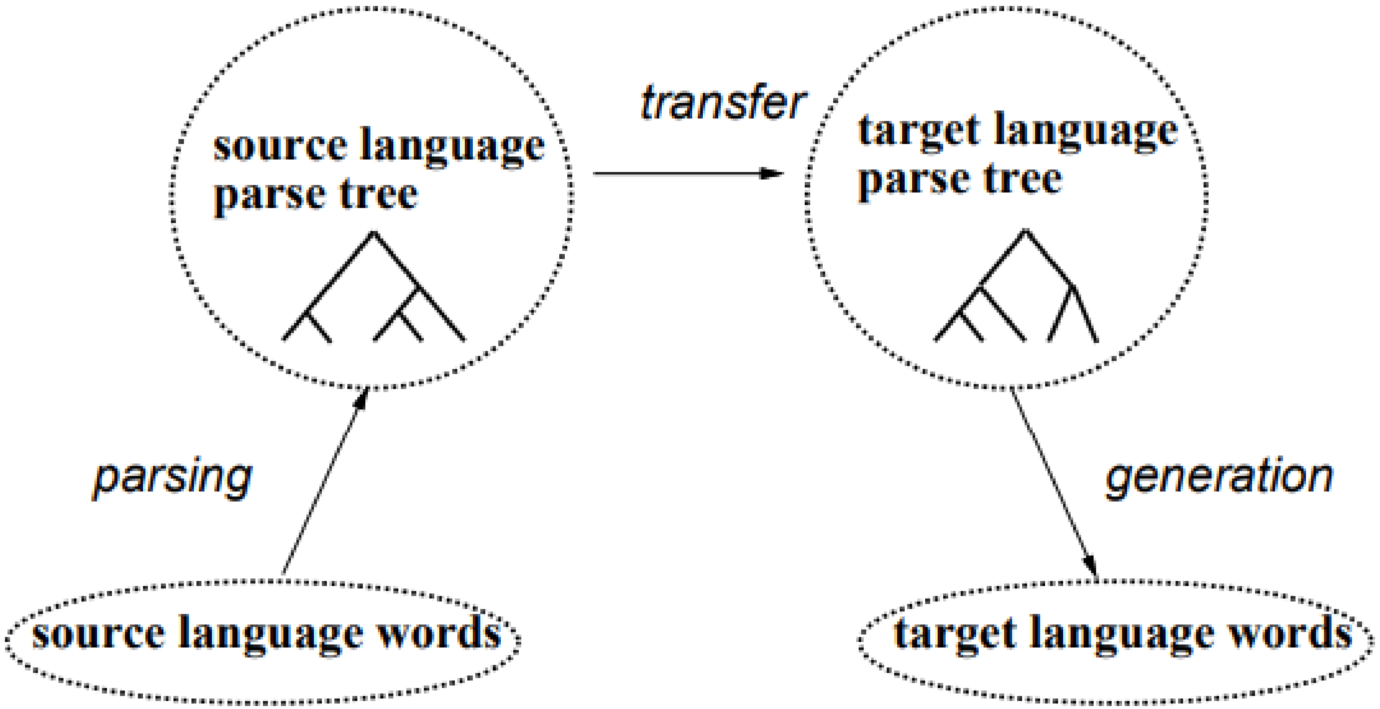
Language Divergence ➔ *the great diversity among languages of the world*

The central problem of MT is to bridge this language divergence

Types of Machine Translation



Transfer architecture for Machine Translation



Language Divergence

Word order: SOV (Hindi), SVO (English), VSO, OSV

S V O

E: Germany won the last World Cup

H: जर्मनी ने पिछला विश्व कप जीता था

S O V

Free (Hindi) vs rigid (English) word order

पिछला विश्व कप जर्मनी ने जीता था *(correct)*

The last World Cup Germany won *(grammatically incorrect)*

The last World Cup won Germany *(meaning changes)*

Language Divergence

Analytic vs Polysynthetic languages

Analytic (Chinese) → very few morphemes per word, no inflections

Polysynthetic (Finnish)→ many morphemes per word, no inflections

English: *Even if it does not rain*

Malayalam: മഴ പെയ്യുതിലെങ്ങിലും

(rain_noun shower_verb+not+even_if+then_also)

Language Divergence

Inflectional systems [infixing (Arabic), fusional (Hindi), agglutinative (Marathi)]

Arabic
k-t-b: root word
katabtu: I wrote
kattabtu: I had (something) written
kitaab: book
kotub: books

Hindi
Jaaunga (1^{st} per, singular, masculine)
Jaaoge (2^{nd} per)
Jaayega (3^{rd} per, singular, masculine)
Jaayenge (3^{rd} per, plural)

Marathi
कपाटावरील: कपाट + वर + ईल
(the one over the cupboard)
दारावरील: दार + वर + ईल
(the one over the door)
दारामागील: दार + मागे + ईल
(the one behind the door)

Language Divergence

<u>Different ways of expressing same concept</u>

water → पानी, जल, नीर

<u>Language registers</u>

Formal: आप बैठिये — Informal: तू बैठ

Standard : मुझे डोसा चाहिए — Dakhini: मेरे को डोसा होना

Language Divergence

- Case marking systems
- Categorical divergence
- Null Subject Divergence
- Pleonastic Divergence

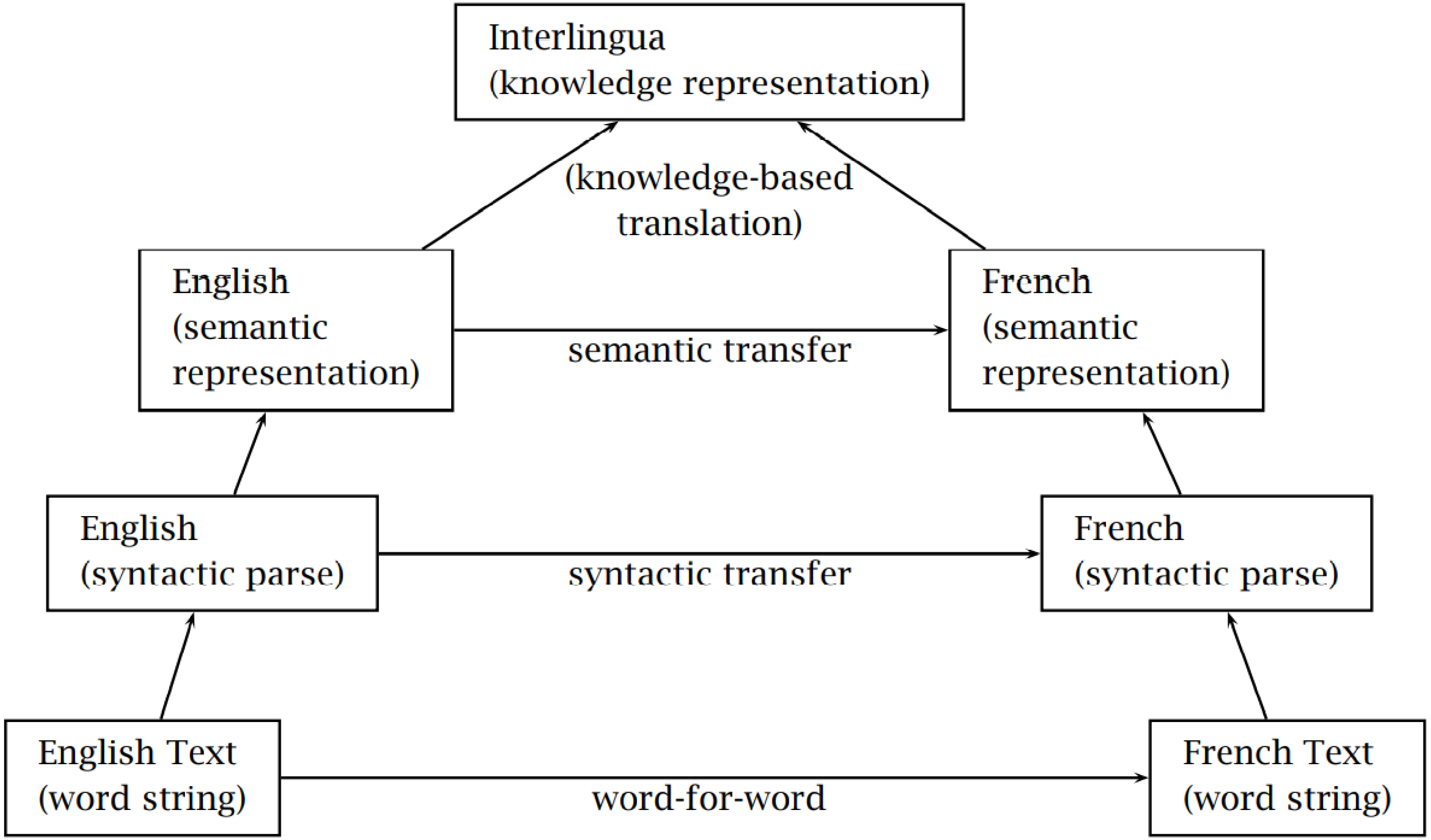
… and much more

Why is Machine Translation difficult?

- **Ambiguity**
  - Same word, multiple meanings: मंत्री (minister or chess piece)
  - Same meaning, multiple words: जल, पानी, नीर (water)
- **Word Order**
  - Underlying deeper syntactic structure
  - Phrase structure grammar?
  - Computationally intensive
- **Morphological Richness**
  - Identifying basic units of words

Different strategies for Machine Translation

- Examples : English (the source) to French (the target).
- Word-based methods translate the source word by word.
- Transfer methods build a structured representation of the source (syntactic or semantic), transform it into a structured representation of the target and generate a target string from this representation.
- Semantic methods use a richer semantic representation than parse trees with, for example, quantifier scope disambiguated.
- Interlingua methods translate via a language-independent knowledge representation


Interlingua
(knowledge representation)
(knowledge-based
translation)
English
(semantic
representation)
semantic transfer
French
(semantic
representation)
English
(syntactic parse)
syntactic transfer
French
(syntactic parse)
English Text
(word string)
word-for-word
French Text
(word string)

Basic Issues in Machine Translation

- Machine translation (MT) is a hard problem
- The goal of many NLP researchers is instead to produce close to error-free output.
- ***word for word:*** One problem - there is no one-to-one correspondence between words in different languages.
  - Lexical ambiguity is one reason
  - One needs to look at context larger than the individual word to choose the correct French translation for ambiguous words like suit.
  - Different word orders

Contd…

- A native word-for-word translation will usually get the word order in the target language wrong.
- This problem is syntactic transfer addressed by the syntactic transfer approach.
- First parse the source approach text, then transform the parse tree of the source text into a syntactic tree in the target language (using appropriate rules), and then generate the translation from this syntactic tree.
- Again faced with ambiguity, syntactic ambiguity - assuming that we can correctly disambiguate the source text.

Contd…

- The syntactic transfer approach solves problems of word order but of ten a syntactically correct translation has inappropriate semantics.
- Eg. In German Ich esse gern ‘I like to eat’ is a verb-adverb construction that translates literally as I eat readily (or willingly, with pleasure, gladly).
- There is no verb-adverb construction in English that can be used to express the meaning of I like to eat.
- A syntax-based approach cannot work here.

Contd…

- In semantic transfer approaches - represents the meaning of the approach source sentence.
- Then generate the translation from the meaning.
- This will fix cases of syntactic mismatch, but even this is not general enough to work for all cases.
- The reason is that even if the literal meaning of a translation is correct, it can still be unnatural to the point of being unintelligible.

Contd…

- A classic example is the way that English and Spanish express direction and manner of motion.
- In Spanish, the direction is expressed using the verb and the manner is expressed with a separate phrase:

La botella entró a la cueva flotando.

- English speakers may be able to understand the literal translation

The bottle entered the cave floating

Contd…

- But if there are too many such literal translations in a text, then it becomes cumbersome to read.
- The correct English translation expresses the manner of motion using the verb and the direction using a preposition:

The bottle floated into the cave

Interlingua

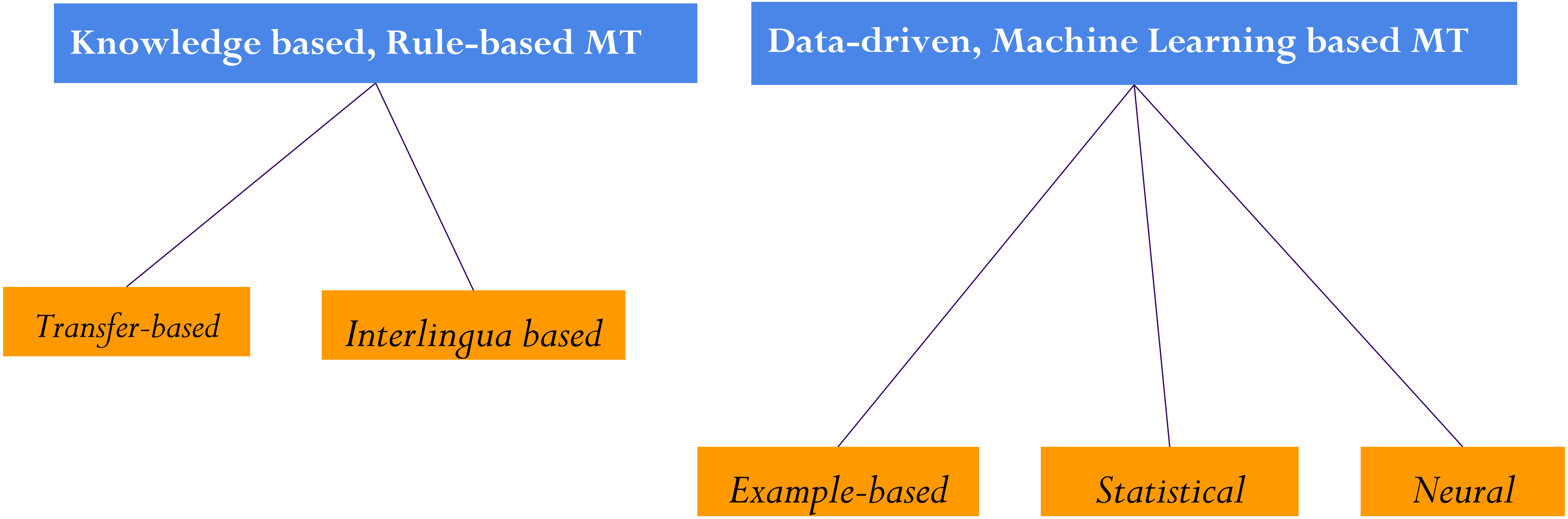
- An interlingua is a knowledge representation formalism that is independent of the way particular languages express meaning.
- An interlingua has the added advantage that it efficiently addresses the problem of translating for a large number of languages, as is, for example, necessary in the European Community.
- Instead of building O(n2) translation systems, for all possible pairs of languages, one only has to build O(n) systems to translate between each language and the interlingua.

Interlingua

- Despite these advantages, there are significant practical problems with interlingua approaches due to the difficulty of designing efficient
- Comprehensive knowledge representation formalisms and due to the large amount of ambiguity that has to be resolved to translate from a natural language to a knowledge representation language.

Machine Translation Paradigm

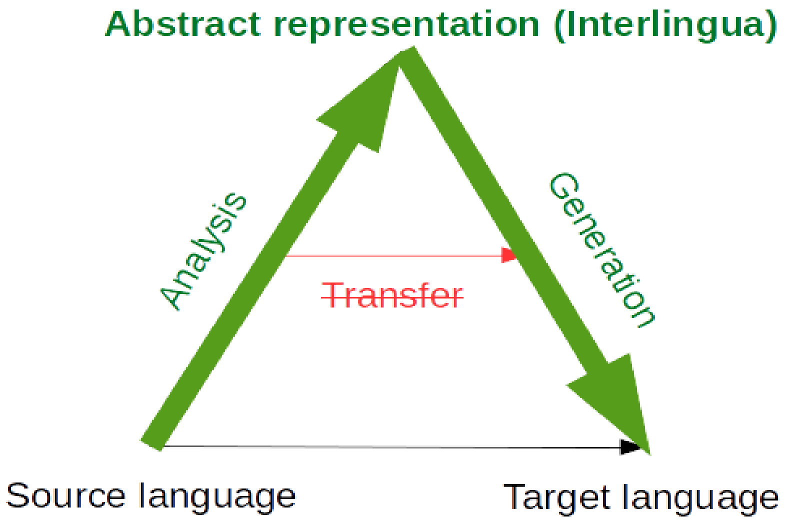
Approaches to build MT systems



Rule-based MT

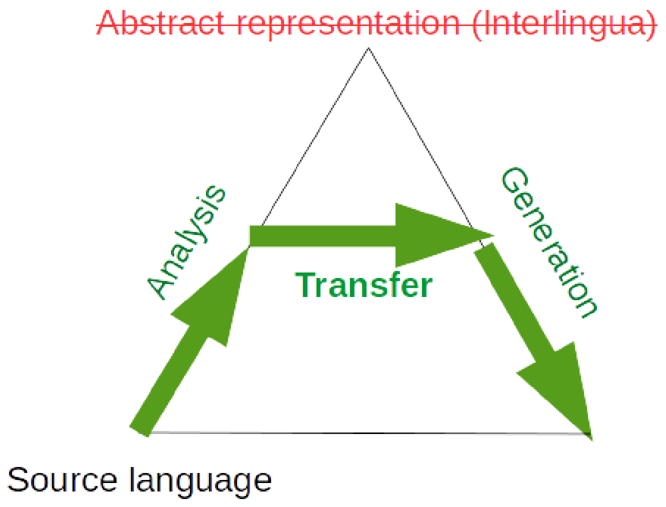
- Rules are written by ***linguistic experts*** to analyze the source, generate an intermediate representation, and generate the target sentence
- Depending on the depth of analysis: interlingua or transfer-based MT

Interlingua based MT



Deep analysis, complete disambiguation and language independent representation

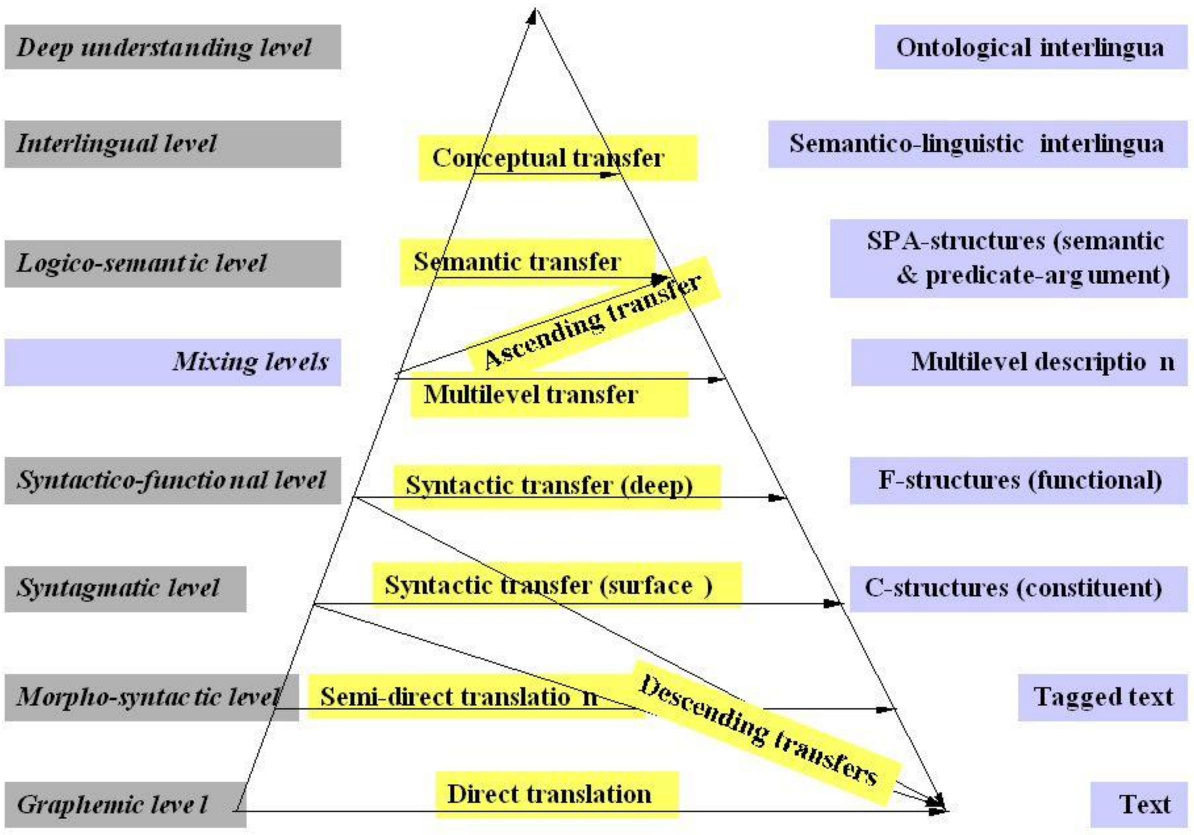
Transfer based MT



Partial analysis, partial disambiguation and a bridge intermediate representation

Vauquois Triangle

Translation approaches can be classified by the depth of linguistic analysis they perform

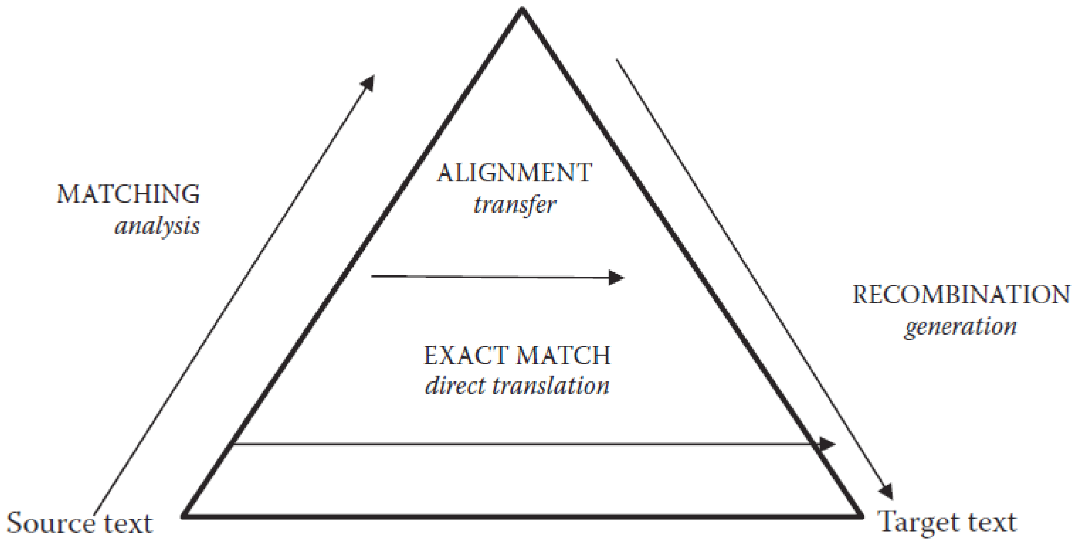


Problems with rule-based MT

- Required linguistic expertise to develop systems
- Maintenance of system is difficult
- Difficult to handle ambiguity
- Scaling to a large number of language pairs is not easy

Example-based MT

Translation by analogy ⇒ *match parts of sentences to known translations and then combine*

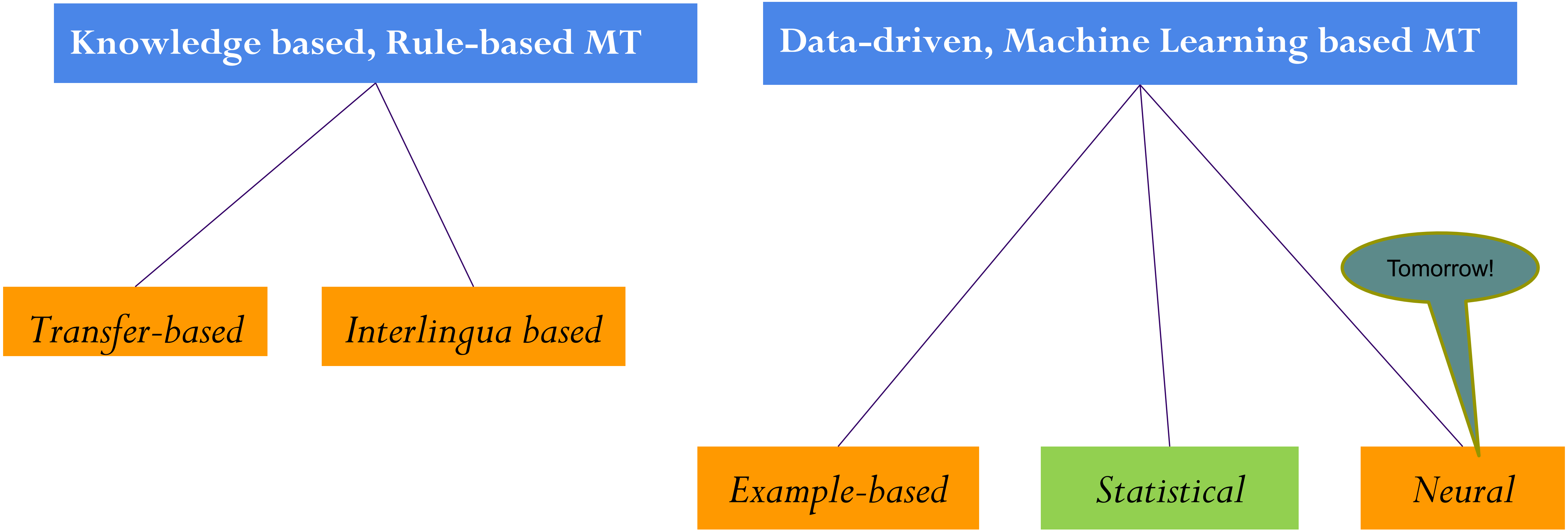


Input: *He buys a book on international politics*

1. **Phrase fragment matching: (*data-driven*)**
   he buys a book international politics
2. **Translation of segments: (*data-driven*)**
   वह खरीदता है एक किताब
   अंतर राष्ट्रीय राजनीति
3. **Recombination:** *(human crafted rules/templates)*
   वह अंतर राष्ट्रीय राजनीति पर एक किताब खरीदता है

- *Partly rule-based, partly data-driven.*
- *Good methods for matching and large corpora did not exist when proposed*

Approaches to build MT systems



A Probabilistic Formalism

STATISTICAL MACHINE TRANSLATION

Let’s formalize the translation process

*We will model translation using a **probabilistic model**. Why?*

- *We would like to have a measure of confidence for the translations we learn*
- *We would like to model uncertainty in translation*

E: target language *e: source language sentence*

F: source language *f : target language sentence*

Best translation

$$\bar{e} = \arg\max_{e} P(e|f)$$

How do we **model** this quantity?

Model: *a simplified and idealized understanding of a physical process*

We must first explain the process of translation

We explain translation using the Noisy Channel Model

A very general framework for many NLP problems

Generate target sentence

Channel corrupts the target

Source sentence is a corruption of the target sentence

Language Model (LM)
P(e)

Captures fluency

Translation Model (TM)
P(f|e)

Captures fidelity

$$P(e|f) = P(e) \times P(f|e)$$

Translation is the process of recovering the original signal given the corrupted signal

Why use this counter-intuitive way of explaining translation?

- Makes it easier to mathematically represent translation and learn probabilities
- **Fidelity** and **Fluency** can be modelled separately

word alignment

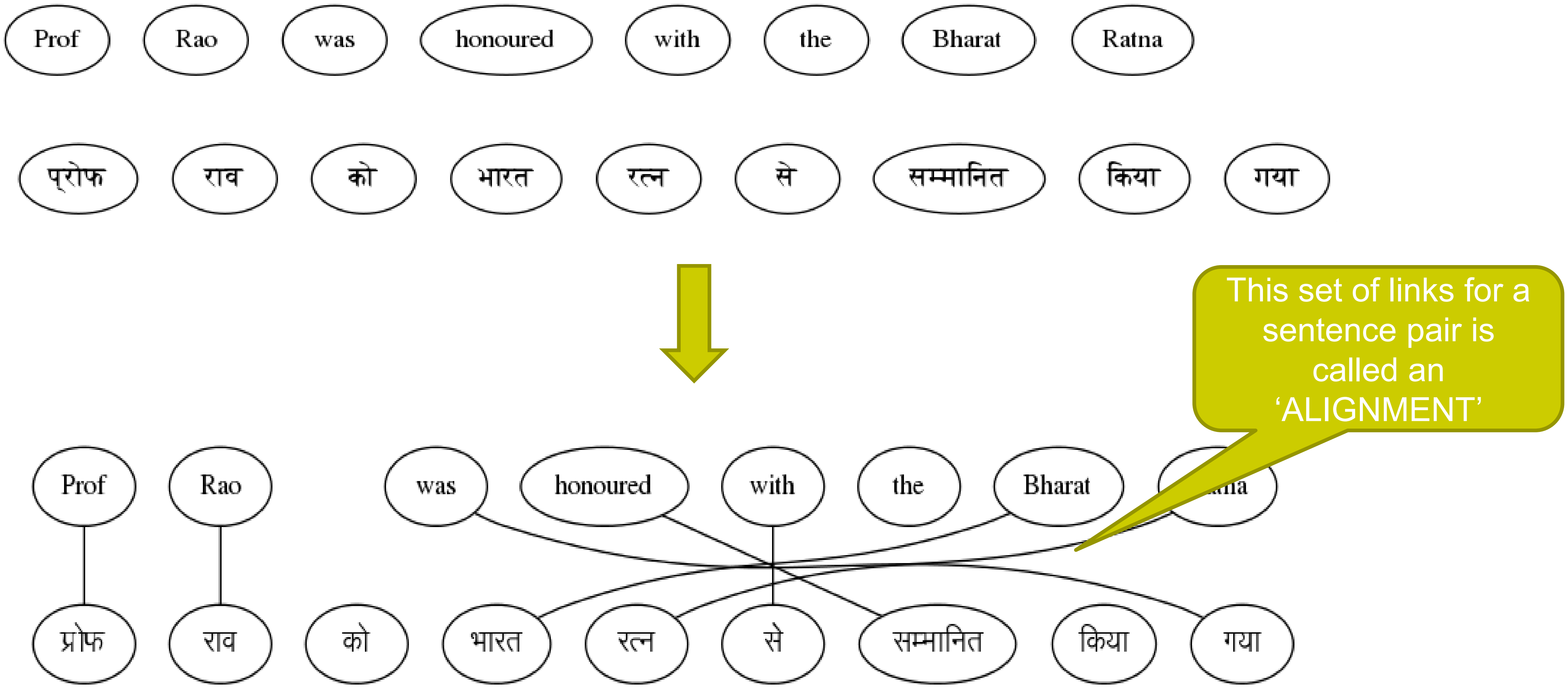
Word Alignment

- A common use of aligned texts is the derivation of bilingual dictionaries dictionaries and terminology databases.

This is usually done in two steps.

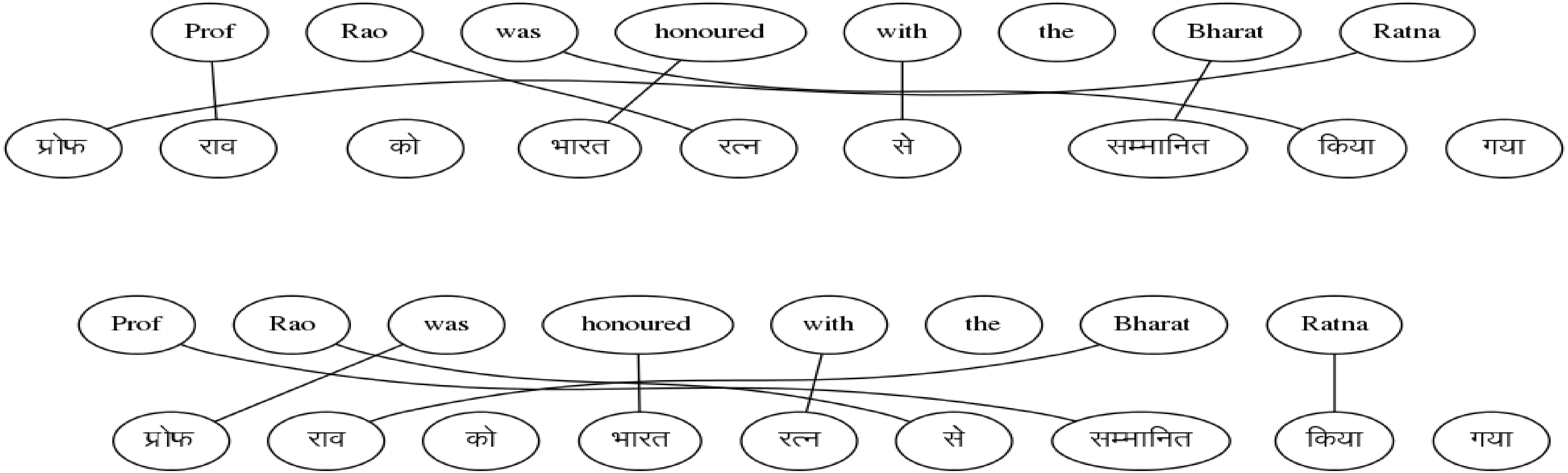
- First the terminology databases text alignment is extended to a word alignment (unless we are dealing with an approach in which word and text alignment are induced simultaneously).
- Then some criteria such as frequency is used to select aligned.

Given a parallel sentence pair, find word level correspondences



But there are multiple possible alignments

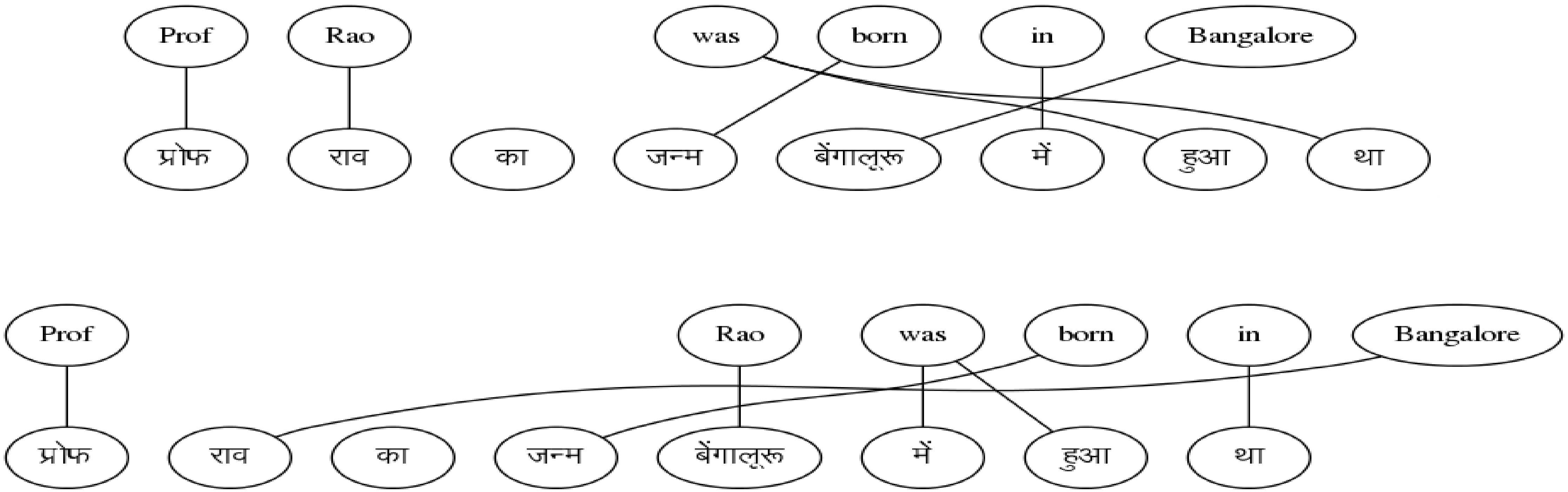
Sentence 1



With one sentence pair, we cannot find the correct alignment

Can we find alignments if we have multiple sentence pairs?

Sentence 2



Yes, let's see how to do that …

Parallel Corpus	
A boy is sitting in the kitchen	एक लडका रसोई मे बैठा है
A boy is playing tennis	एक लडका टेनिस खेल रहा है
A boy is sitting on a round table	एक लडका एक गोल मेज पर बैठा है
Some men are watching tennis	कुछ आदमी टेनिस देख रहे है
A girl is holding a black book	एक लडकी ने एक काली किताब पकडी है
Two men are watching a movie	दो आदमी चलचित्र देख रहे है
A woman is reading a book	एक औरत एक किताब पढ रही है
A woman is sitting in a red car	एक औरत एक काले कार मे बैठी है

Parallel Corpus	
A boy is **sitting** in the kitchen	एक लडका रसोई मे **बैठा** है
A boy is playing **tennis**	एक लडका **टेनिस** खेल रहा है
A boy is **sitting** on a round table	एक लडका एक गोल मेज पर **बैठा** है
Some men **are watching tennis**	कुछ आदमी **टेनिस देख रहे है**
A girl is holding a black book	एक लडकी ने एक काली किताब पकडी है
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A woman is reading a book	एक औरत एक किताब पढ रही है
A woman is **sitting** in a red car	एक औरत एक काले कार मे **बैठा** है

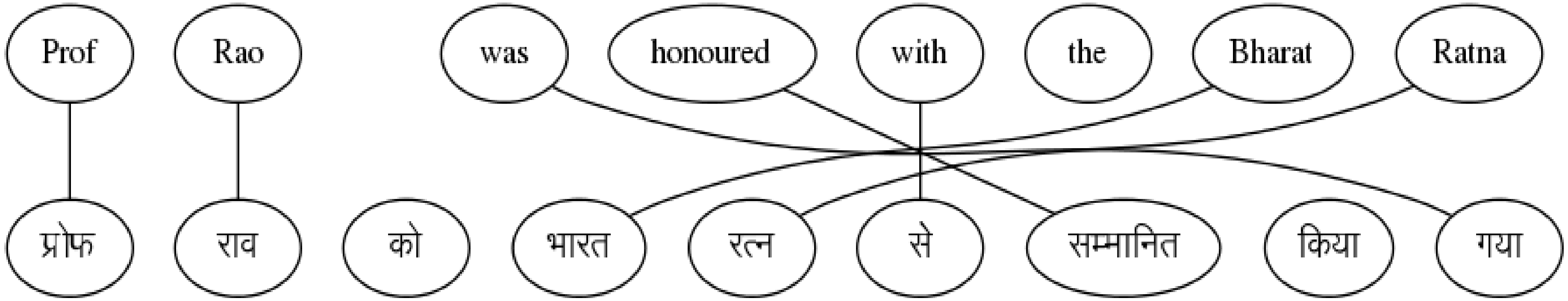
Key Idea

Co-occurrence of translated words

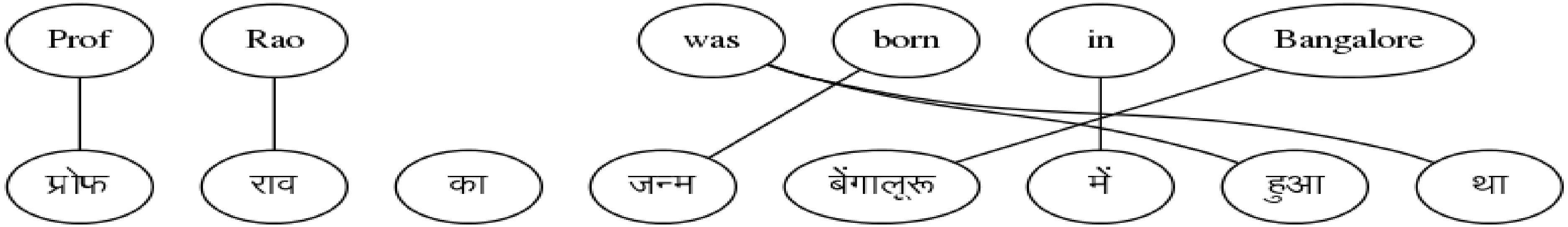
Words which occur together in the parallel sentence are likely to be translations (higher $P(f|e)$)

If we knew the alignments, we could compute P(f|e)

Sentence 1



Sentence 2



$$P(f|e) = \frac{\#(f,e)}{\#(*,e)}$$

$$P(Prof|\text{प्रोफ}) = \frac{2}{2}$$

$\#(a,b)$: *number of times word a is aligned to word b*

But, we can find the best alignment only if we know the word translation probabilities

The best alignment is the one that maximizes the sentence translation probability

$$P(\boldsymbol{f}, \boldsymbol{a}|\boldsymbol{e}) = P(a) \prod_{i=1}^{i=m} P\left(f_i \middle| e_{a_i}\right) \Rightarrow \boldsymbol{a}^* = \underset{\boldsymbol{a}}{\operatorname{argmax}} \prod_{i=1}^{i=m} P\left(f_i \middle| e_{a_i}\right)$$

This is a chicken and egg problem! How do we solve this?

We can solve this problem using a two-step, iterative process

Start with random values for word translation probabilities

Step 1: Estimate alignment probabilities using word translation probabilities

Step 2: Re-estimate word translation probabilities

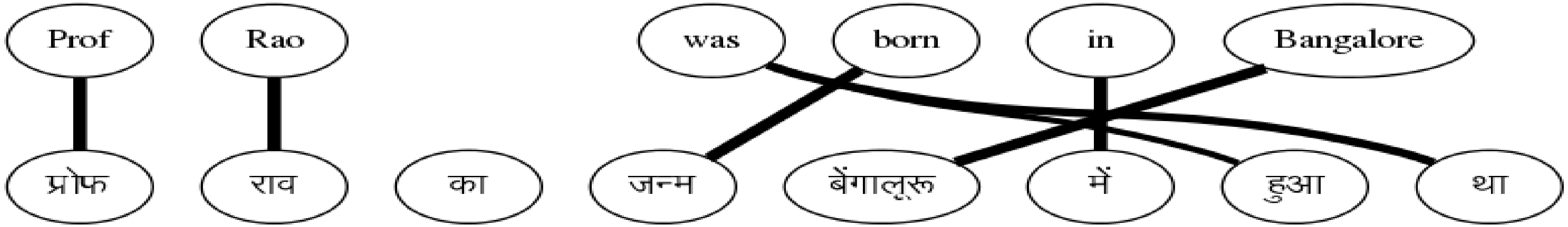
- *We don't know the best alignment*
- *So, we consider all alignments while estimating word translation probabilities*
- *Instead of taking only the best alignment, we consider all alignments and weigh the word alignments with the alignment probabilities*

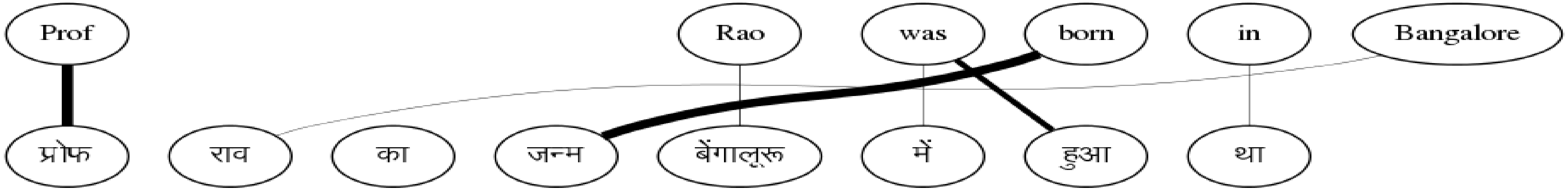
$$P(f|e) = \frac{expected\ \#(f,e)}{expected\ \#(*,e)}$$

Repeat Steps (1) and (2) till the parameters converge

At the end of the process …

Sentence 2





Is the algorithm guaranteed to converge?

That's the nice part → it is guaranteed to converge

This is an example of the well known Expectation-Maximization Algorithm

However, the problem is highly non-convex

Will lead to local minima

Good modelling assumptions necessary to ensure a good solution

Summary

- EM provides a semi-supervised method for learning word alignments and word translation probabilities
- Word translation probabilities can be used to extract a bilingual dictionary Avoids the new for word-aligned corpora
- If a few word-aligned sentences are available, discriminative alignment methods can improve upon the EM-based solution
  - Arbitrary features can be incorporated
  - Morphological information
  - Character level edit distance

Phrase Based Translation

What is PB-SMT?

Why stop at learning word correspondences?

KEY IDEA → Use "Phrase" (Sequence of Words) as the basic translation unit

Note: the term 'phrase' is not used in a linguistic sense

The Prime Minister of India	भारत के प्रधान मंत्री bhArata ke pradhAna maMtrI India of Prime Minister
is running fast	तेज भाग रहा है teja bhAg rahA hai fast run -continuous is
honoured with	से सम्मानित किया se sammanita kiyA with honoured did
Rahul lost the match	राहुल मुकाबला हार गया rAhula mukAbalA hAra gayA Rahul match lost

Parallel Corpus	
A boy is **sitting** in the kitchen	एक लडका रसोंई मे **बैठा** है
A boy is playing **tennis**	एक लडका **टेनिस** खेल रहा है
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A woman is **sitting** in a red car	एक औरत एक काले कार मे **बैठा** है

Benefits of PB-SMT

Local Reordering → Intra-phrase re-ordering can be memorized

The Prime Minister of India	भारत के प्रधान मंत्री bhaarat ke pradhaan maMtrI India of Prime Minister

Sense disambiguation based on local context → Neighbouring words help make the choice

heads towards Pune	पुणे की ओर जा रहे है pune ki or jaa rahe hai Pune towards go –continuous is
heads the committee	समिति की अध्यक्षता करते है Samiti kii adhyakshata karte hai committee of leading -verbalizer is

Benefits of PB-SMT (2)

Handling institutionalized expressions

- Institutionalized expressions, idioms can be learnt as a single unit

hung assembly	त्रिशंकु विधानसभा trishanku vidhaansabha
Home Minister	गृह मंत्री gruh mantrii
Exit poll	चुनाव बाद सर्वेक्षण chunav baad sarvekshana

Improved Fluency

- The phrases can be arbitrarily long (even entire sentences)

Mathematical Model

$$e_{\text{best}} = \text{argmax}_{\mathbf{e}}\, p(\mathbf{e}|\mathbf{f}) = \text{argmax}_{\mathbf{e}}\, p(\mathbf{f}|\mathbf{e})\, p_{\text{LM}}(\mathbf{e})$$

Let's revisit the decision rule for SMT model

Let's revisit the translation model $p(\mathbf{f}\,|\,\mathbf{e})$

-Source sentence can be segmented in **I** phrases

-Then, $p(\mathbf{f}\,|\,\mathbf{e})$ can be decomposed as:

$$p(\bar{f}_1^I|\bar{e}_1^I) = \prod_{i=1}^{I} \underbrace{\phi(\bar{f}_i|\bar{e}_i)}_{\text{Phrase Translation Probability}}\, \underbrace{d(\text{start}_i - \text{end}_{i-1} - 1)}_{\text{Distortion probability}}$$

start_i :start position in **f** of i^{th} phrase of **e**

end_i :end position in **f** of i^{th} phrase of **e**

Learning The Phrase Translation Model

Involves Structure + Parameter Learning:

Learn the **Phrase Table**: the central data structure in PB-SMT

The Prime Minister of India	भारत के प्रधान मंत्री
is running fast	तेज भाग रहा है
the boy with the telescope	दूरबीन से लड़के को
Rahul lost the match	राहुल मुकाबला हार गया

Learn the **Phrase Translation Probabilities**

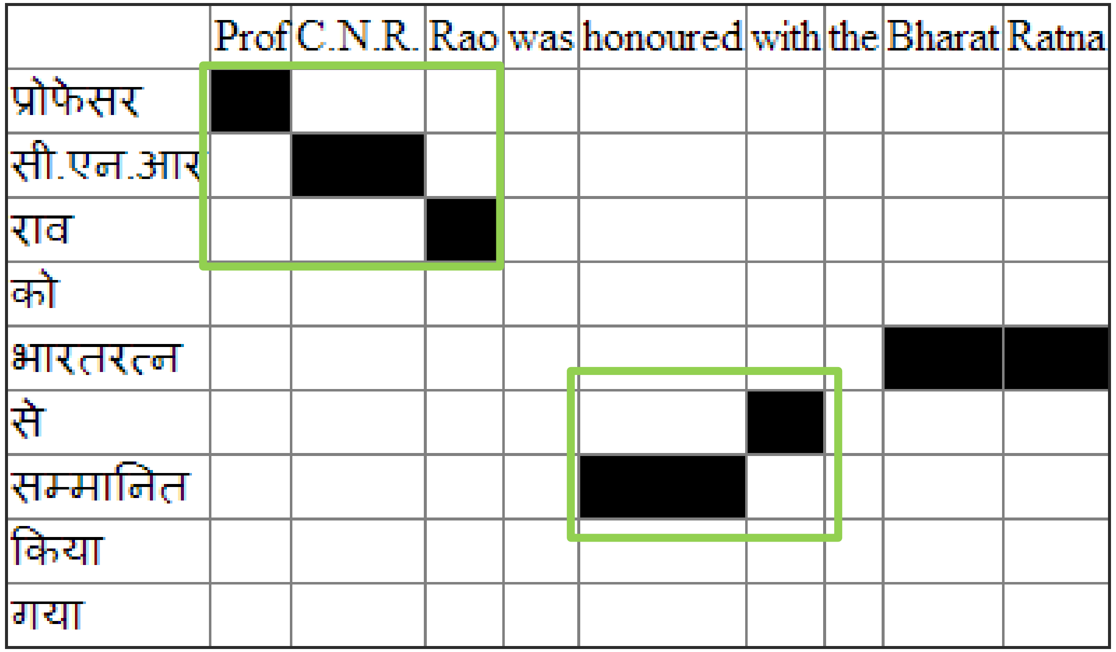
Prime Minister of India	भारत के प्रधान मंत्री India of Prime Minister	0.75
Prime Minister of India	भारत के भूतपूर्व प्रधान मंत्री India of former Prime Minister	0.02
Prime Minister of India	प्रधान मंत्री Prime Minister	0.23

Learning Phrase Tables from Word Alignments

Start with word alignments

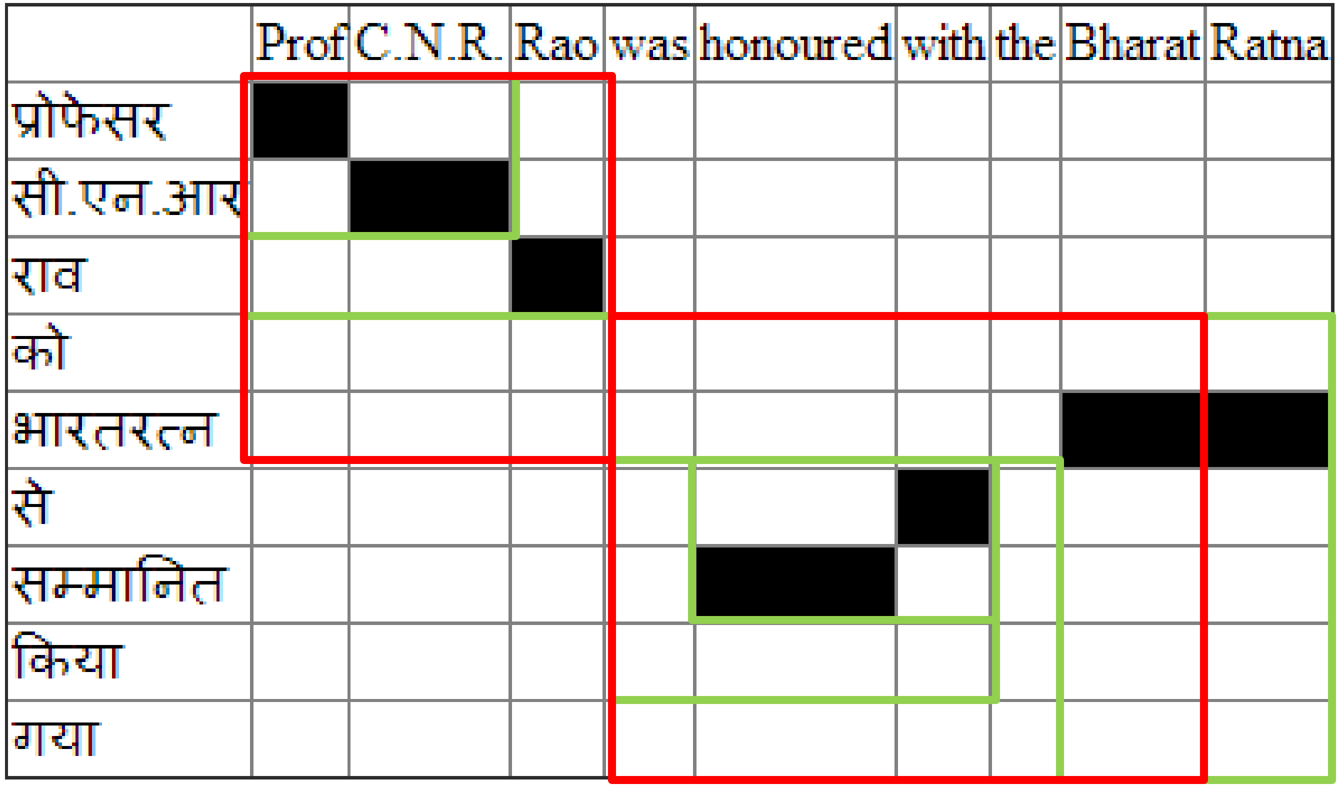
- Word Alignment : reliable input for phrase table learning
  - high accuracy reported for many language pairs

Central Idea: A consecutive sequence of aligned words constitutes a "phrase pair"

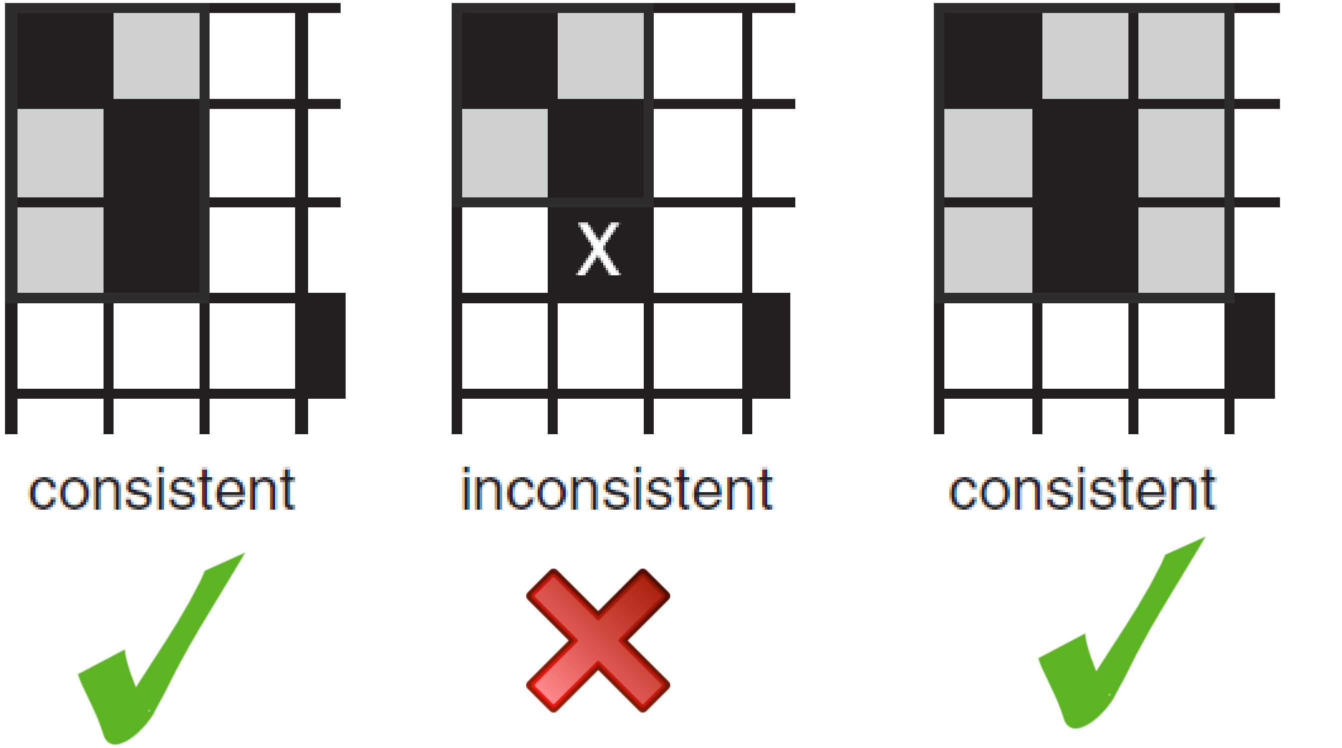


Which phrase pairs to include in the phrase table?

Extracting Phrase Pairs

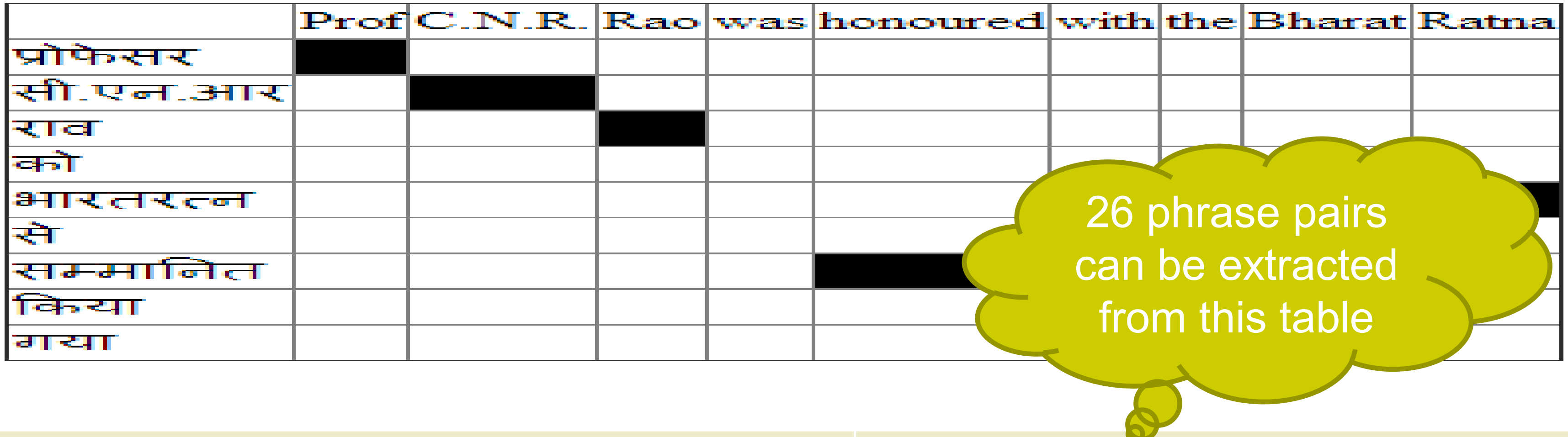


Phrase Pairs “consistent” with word alignment



Source: SMT, Phillip Koehn

Examples



Professor CNR	प्रोफेसर सी.एन.आर
Professor CNR Rao	प्रोफेसर सी.एन.आर राव
Professor CNR Rao was	प्रोफेसर सी.एन.आर राव
Professor CNR Rao was	प्रोफेसर सी.एन.आर राव को
honoured with the Bharat Ratna	भारतरत्न से सम्मानित
honoured with the Bharat Ratna	भारतरत्न से सम्मानित किया
honoured with the Bharat Ratna	भारतरत्न से सम्मानित किया गया
honoured with the Bharat Ratna	को भारतरत्न से सम्मानित किया गया

Computing Phrase Translation Probabilities

- Estimated from the relative frequency:

$$\phi(\bar{f}|\bar{e}) = \frac{\text{count}(\bar{e},\bar{f})}{\sum_{\bar{f}_i} \text{count}(\bar{e},\bar{f}_i)}$$

Prime Minister of India	भारत के प्रधान मंत्री India of Prime Minister	*0.75*
Prime Minister of India	भारत के भूतपूर्व प्रधान मंत्री India of former Prime Minister	*0.02*
Prime Minister of India	प्रधान मंत्री Prime Minister	*0.23*

Discriminative Training of PB-SMT

Directly model the posterior probability p**(e|f)**

Use the Maximum Entropy framework

$$P(\mathbf{e}|\mathbf{f}) = \exp\left(\sum_i \lambda_i h_i(f_1^I, e_1^J)\right)$$

$$e^* = \arg\max_{e_i} \sum_i \lambda_i h_i(f_1^I, e_1^J)$$

- h_i**(f,e)** are feature functions , λ_i's are feature weights

Benefits:

- Can add arbitrary features to score the translations
- Can assign different weight for each features
- Assumptions of generative model may be incorrect

More features for PB-SMT

Inverse phrase translation probability ($\phi(\bar{f}|\bar{e})$)

Lexical Weighting

$$\text{lex}(\bar{e}|\bar{f}, a) = \prod_{i=1}^{\text{length}(\bar{e})} \frac{1}{|\{j|(i,j) \in a\}|} \sum_{\forall (i,j) \in a} w(e_i|f_j)$$

- *a:* alignment between words in phrase pair (ē, f)
- *w(x|y):* word translation probability

Inverse Lexical Weighting

- Same as above, in the other direction

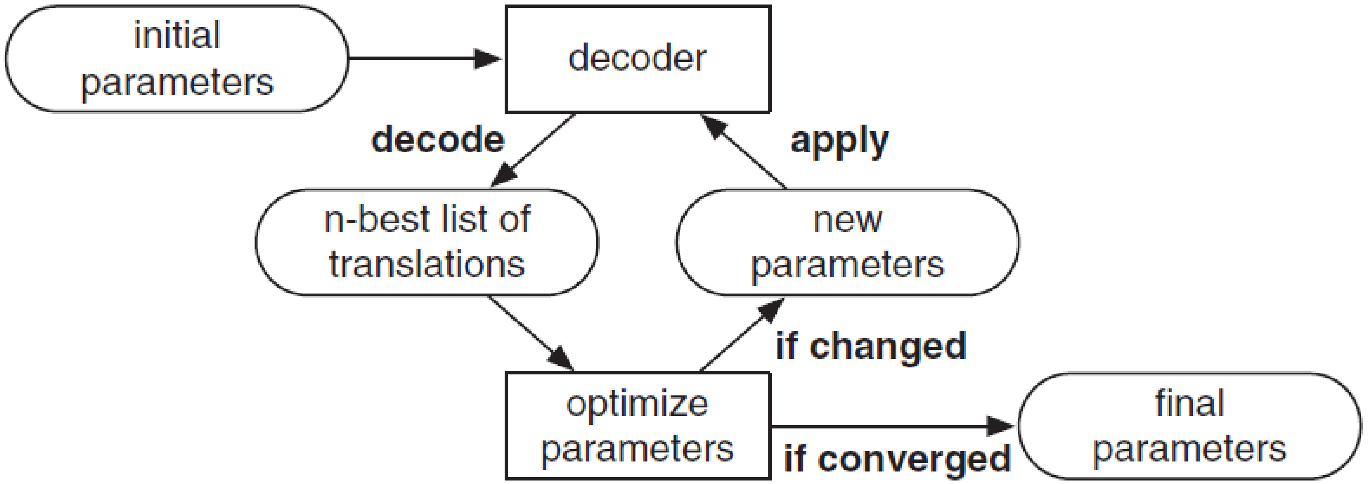
Tuning

Learning feature weights from data – λ_i

Minimum Error Rate Training (MERT)

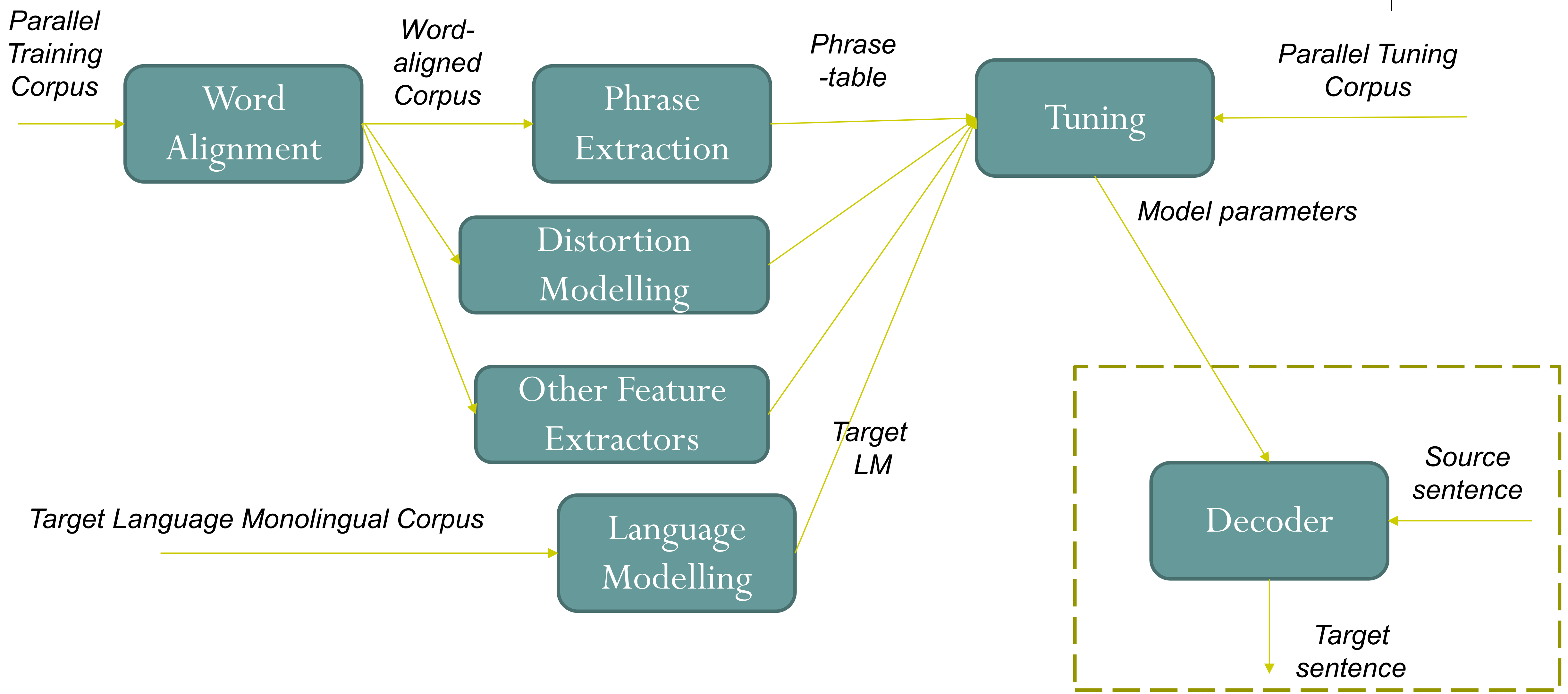
Search for weights which minimize the translation error on a held-out set (tuning set)

- Translation error metric : (1 – *BLEU)*



Source: SMT, Phillip Koehn

Typical SMT Pipeline

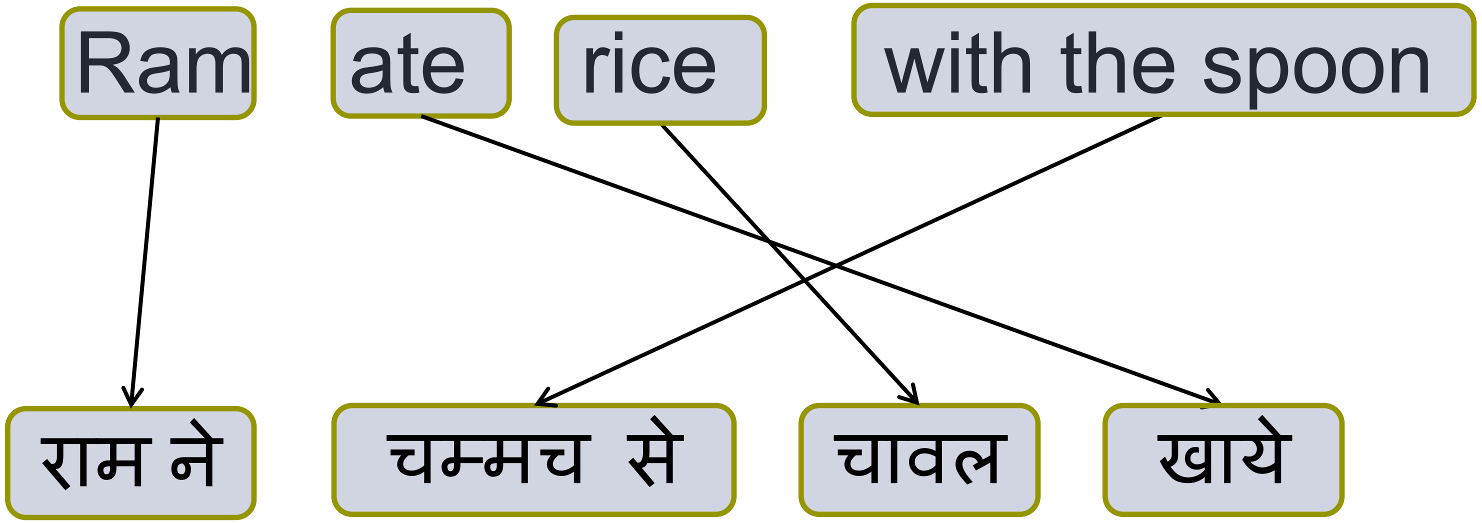


Decoding

Searching for the best translations in the space of all translations

$$\boxed{e^* = \arg\max_{e_i} \sum_i \lambda_i h_i(f_1^I, e_1^J)}$$

An Example of Translation

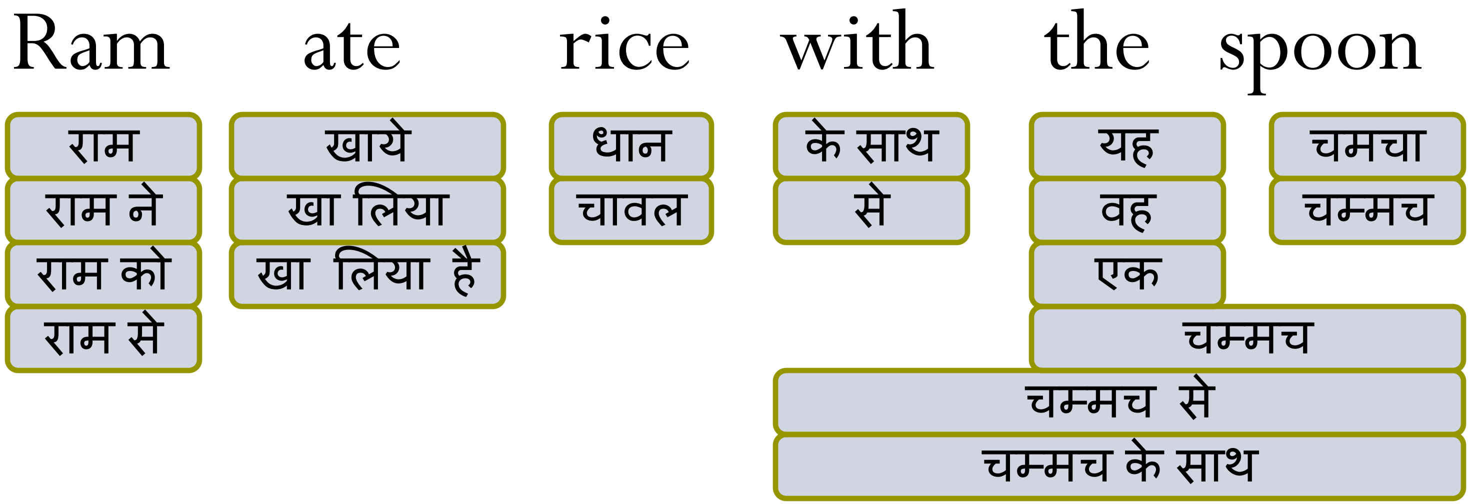


Reality

We picked the phrase translation that made sense to us

The computer has less intuition

Phrase table may give many options to translate the input sentence



What is the challenge in decoding?

The task of decoding in machine translation is to find the best scoring translation according to translation models

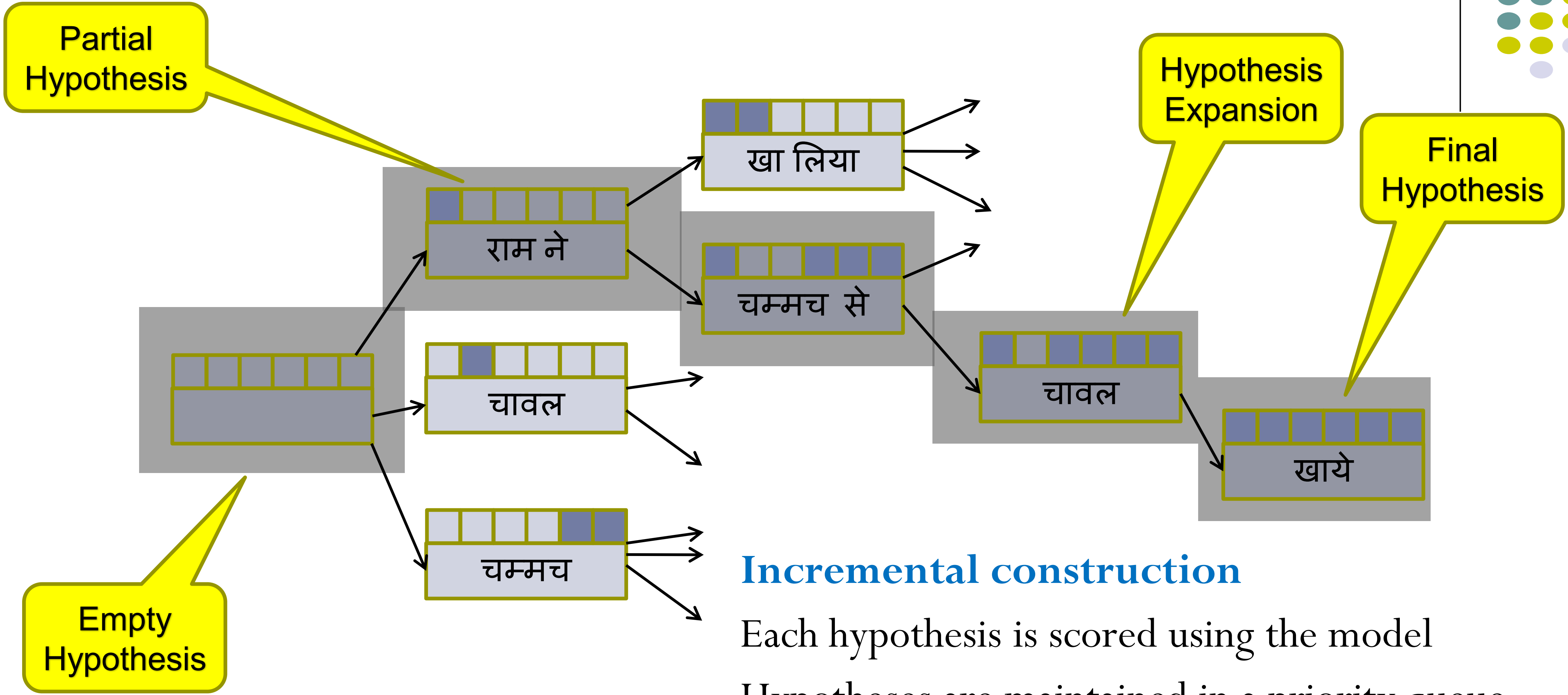
Hard problem, since there is a exponential number of choices, given a specific input sentence

Shown as an NP complete problem

Need to come up with heuristic search methods

No guarantee of finding the best translation

Search Space and Search Organization



Incremental construction

Each hypothesis is scored using the model

Hypotheses are maintained in a priority queue

Limit to the reordering window for efficiency

EXTENSIONS TO PHRASE-BASED SMT

- Addressing Word-order Divergence
- Addressing Morphological Divergence
- Handling Named Entities

We have looked at a basic phrase-based SMT system

This system can learn word and phrase translations from parallel corpora

But many important linguistic phenomena need to be handled

- **Divergent Word Order**
- Rich morphology
- Named Entities and Out-of-Vocabulary words

GETTING WORD ORDER RIGHT

Phrase based MT is not good at learning word ordering

Solution: Let's help PB-SMT with some preprocessing of the input
Change order of words in input sentence to match order of the words in the target language

Let's take an example

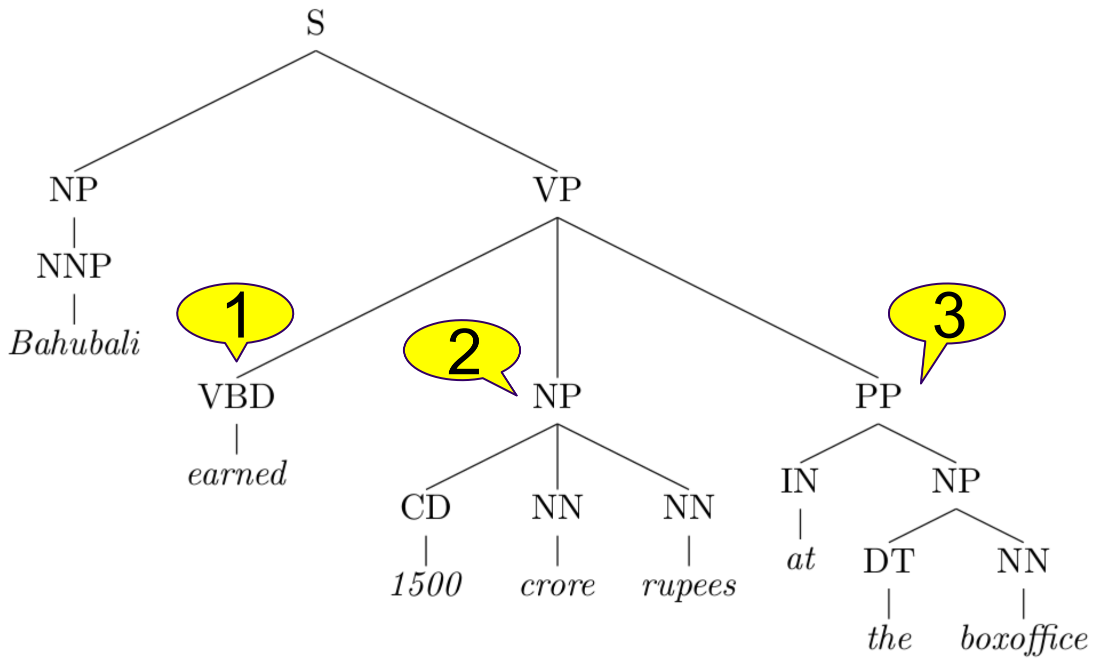
Bahubali earned more than 1500 crore rupee sat the boxoffice

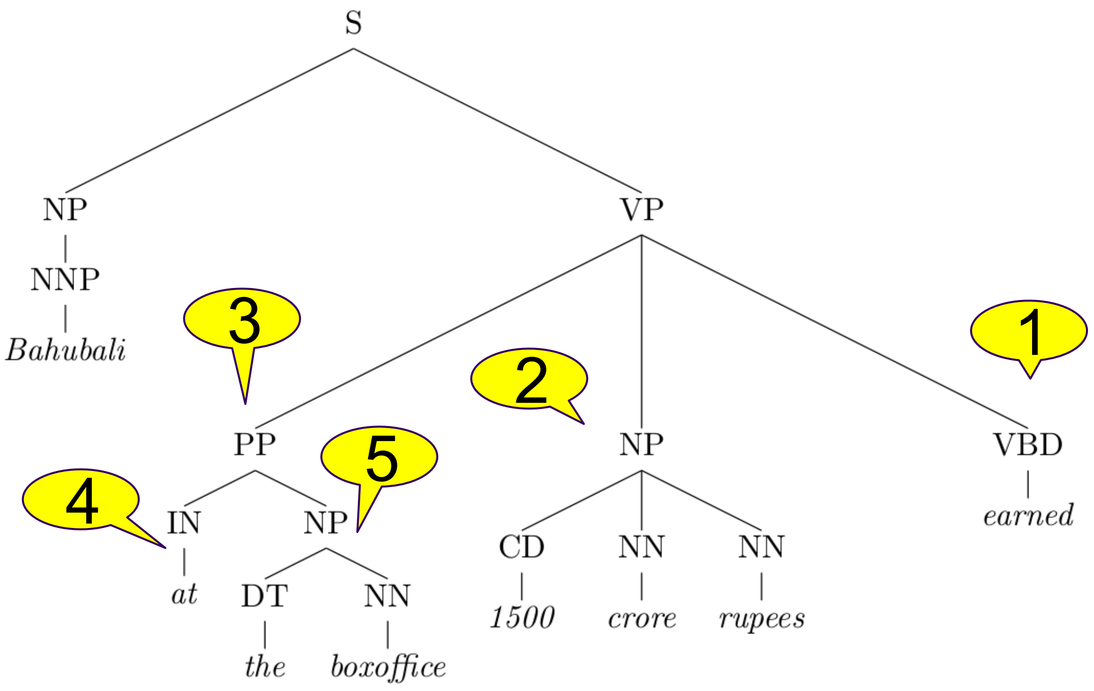
Parse the sentence to understand its syntactic structure

Apply rules to transform the tree

VP → VBD NP PP ⇒ VP → PP NP VBD

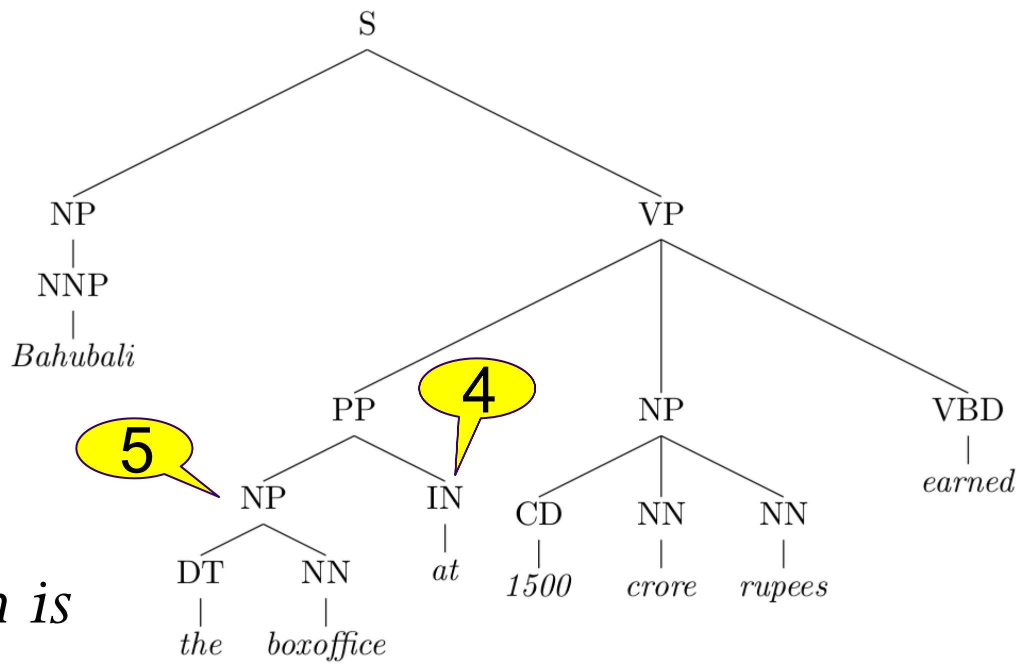
This rule captures Subject-Verb-Object to Subject-Object-Verb divergence





Prepositions in English become postpositions in Hindi

PP ⟶ IN NP ⇒ PP ⟶ NP IN



The new input to the machine translation system is

Bahubali the boxoffice at 1500 crore rupees earned

Now we can translate with little reordering

बाहुबली ने बॉक्सओफिस पर 1500 करोड रुपए कमाए

These rules can be written manually or learnt from parse trees

Better methods exist for generating the correct word order

Incorporate learning of reordering is built into the SMT system

Hierarchical PBSMT ⇒ Provision in the phrase table for limited & simple reordering rules

Syntax-based SMT ⇒ Another SMT paradigm, where the system learns mappings of "treelets" instead of mappings of phrases

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- **Rich morphology**
- Named Entities and Out-of-Vocabulary words

Language is very productive, you can combine words to generate new words

Inflectional forms of the Marathi word **घर**

घर	house
घरात	in the house
घरावरती	on the house
घराखाली	below the house
घरामध्ये	in the house
घरामागे	behind the house
घराचा	of the house
घरामागचा	that which is behind the house
घरासमोर	in front of the house
घरासमोरचा	that which is in front of the house
घरांसमोर	in front of the houses

Hindi words with the suffix **वाद**

साम्यवाद	communism
समाजवाद	socialism
पूंजीवाद	capitalism
जातीवाद	casteism
साम्राज्यवाद	imperialism

The corpus should contains all variants to learn translations

This is infeasible!

Language is very productive, you can combine words to generate new words

Inflectional forms of the Marathi word घर

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घर ◌ा वरती	on the house
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घर ◌ा मागे	behind the house
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साम्राज्य वाद	imperialism

- *Break the words into its component morphemes*
- *Learn translations for the morphemes*
- *Far more likely to find morphemes in the corpus*

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But many important linguistic phenomena need to be handled

- Divergent Word Order
- Rich morphology
- **Named Entities and Out-of-Vocabulary words**

Some words not seen during train will be seen at test time

These are out-of-vocabulary (OOV) words

***Names** are one of the most important category of OOVs*

⇒ There will always be names not seen during training

How do we translate names like Sachin Tendulkar to Hindi?

What we want to do is map the Roman characters to Devanagari to they sound the same when read → सचिन तेंदुलकर

➔ *We call this process '**transliteration**'*

How do we transliterate?

Convert a sequence of characters in one script to another script

s a c h i n → स च ि न

Isn't that a translation problem → at the character level?

Albeit a simpler one,

- *Smaller vocabulary*
- *No reordering*
- *Shorter segments*